Lecture #6: Continuous Probability Distributions

MATH 145 – Calculus I

Torin Quinlivan

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Recall: Axioms of Probability

Definition: A **probability space** is a triple $(\Omega, \mathcal{F}, P)$, where Ω is the **sample space**, $\mathcal{F}$ is the **event space**, and P is a **probability law (or measure)**.

Definition: A **probability law** is a function $P : \mathcal{F} \to [0, 1]$ that maps an event A to a real number in $[0, 1]$. The function must satisfy the **axioms of probability**:

1. **Non-Negativity:** $P(A) \geq 0$, for any $A \subseteq \Omega$.
2. **Normalization:** $P(\Omega) = 1$.
3. **Additivity:** For any disjoint sets $\{A_1, A_2, A_3, \ldots\}$, it must be true that
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Models of Probability Laws

We have two models for a probability law:

1. Counting: If $A \subseteq \Omega$, then $P(A) = \dfrac{|A|}{|\Omega|}$.

2. Probability Mass Functions of Discrete RVs: $X : \Omega \to [R]$.

Theorem: A PMF should satisfy the condition that:

$$\sum_{x \in X(\Omega)} p_X(x) = 1$$

Let's now generalize this to continuous random variables.

Probability Density Functions

A **prbability density function** f_X of a random variable X is a mapping $f_X : \Omega \to \mathbb{R}$, with the properties

1. **Non-Negativity:** $f_X(x) \geq 0$ for all $x \in \Omega$.
2. **Unity:** $\displaystyle\int_\Omega f_X(x)dx = 1$.
3. **Measure of a Set:** $P\Big[\{x \in A\}\Big] = \displaystyle\int_A f_X(x)dx$.

Probability Density Functions

Any function $f(x)$ satisfying these properties defines a probability law using integration.

Let X be a continuous random variable. The **probability density function of X** is a funtion $f_X : \Omega \to \mathbb{R}$ that, when integrated over an interval $[a, b]$, yields the probability of obtaining $a \leq X \leq b$:

$$P[a \leq X \leq b] = \int_a^b f_X(x)dx.$$

The probability density function uniquely defines a random variable. If X and Y share a domain Ω, and $f_X(t) = f_Y(t)$ for all $t \in \Omega$, then $X = Y$.

What is the probability of $P(X = a)$?

Example: Probability Density Function

Example: Let $f_X(x) = 3x^2$ with $\Omega = [0, 1]$.

1. Explain why f_X is a PDF.

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1. Explain why f_X is a PDF.
2. If $A = [0, 0.5]$, find $P(X \in A)$.
3. What does this mean graphically?
4. What is $P(0 \leq X < 0.5)$?

Cumulative Distribution Function

Let X be a continuous random variable with a sample space $\Omega = \mathbb{R}$. The **cumulative distribution function (CDF) of X** is:

$$F_X(x) \stackrel{def}{=} P[X \leq x] = \int_{-\infty}^{x} f_X(t)dt.$$

Let X be a continuous random variable. If the CDF F_X is continuous at any $x \in [a, b]$, then:

$$P[a \leq X \leq b] = F_x(b) - F_X(a)$$

The **survival function of X** is:

$$S(x) = 1 - F_X(x) \stackrel{def}{=} P[X > x] = \int_{x}^{\infty} f_X(t)\, dt.$$

The Cumulative Distribution Function vs. Probability Density Function

The **probability density function (PDF)** is the derivative of the cumulative distribuion function (CDF):

$$f_X(x) = \frac{dF_X(x)}{dx} = \frac{d}{dx}\int_{-\infty}^{x} f_X(t)\,dt,$$

provided F_X is differentiable at x.

If F_x is not differentiable at $x = x_0$, then

$$f_X(x_0) = P[X - x_0]\delta(x - x_0).$$

Example: Cumulative Density Function

Example: Let $f_X(x) = 3x^2$ with $\Omega = [0, 1]$.

5. What is the CDF $F_X(x)$ of X?

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6. Find $P\left(X \leq \frac{1}{4}\right)$ with the PDF $f_X(x)$.

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7. Find $P\left(X \leq \frac{1}{4}\right)$ with the CDF $F_X(x)$.

Properties of the Cumulative Distribution Function

Theorem: Let X be a random variable (either continuous or discrete), then the CDF of X has the following properties:

1. The CDF is **nondecreasing**.
2. The **maximum** of the CDF is when $x = \infty$: $F_X(\infty) = 1$.
3. The **minimum** of the CDF is when $x = -\infty$: $F_x(-\infty) = 0$.

Continuity of the Cumulative Distribution Function

Theorem: For any random variable X (discrete or continuous), $F_X(x)$ is always **right-continuous**. That is,

$$F_X(b) = F_X(b^+) \overset{def}{=} \lim_{h \to 0} F_X(b+h).$$

Theorem: For any random variable X (discrete or continuous), $P(X = b)$ is:

$$P[X = b] = \begin{cases} F_X(b) - F_X(b^-), & \text{if } F_X \text{ is discontinuous ot } x = b, \\ 0, & \text{otherwise.} \end{cases}$$

The Expected Value of a Continuous Random Variable

We can generalize the definition of the expected value we developed with discrete random variables and sums to continuous random variables and integrals.

The **expectation** of a continuous random variable X is:

$$E[X] = \int_{\Omega} x f_X(x)\, dx.$$

Theorem: Let $g : \Omega \to \mathbb{R}$ be a function and X be a continuous random variable. Then:

$$E\Big[g(X)\Big] = \int_{\Omega} g(x) f_X(x)\, dx.$$

The n^th Moment of a Continuous Random Variable

The n^{th} **moment** of a continuous random variable X is

$$E[X^n] = \int_\Omega x^n f_X(x)\, dx.$$

The n^{th} **central moment** of a continuous random variable X is

$$E[(X-\mu)^n] = \int_\Omega (x-\mu)^n f_X(x)\, dx.$$

The Variance of a Continuous Random Variable

The same thing applies to the variance.

The **variance** of a continuous random variable X is:

$$V[X] = \int_{\Omega} (x-\mu)^2 f_X(x)\, dx.$$

The formulas concerning the variance and the expected value hold, such as $V(X) = E(X^2) - [E(X)]^2$.

Example: Expectation and Variance

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10. What is the expected value of $4X + 9$, $E(4X + 9)$?

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10. What is the expected value of $4X + 9$, $E(4X + 9)$?
11. What is the variance of X, $V(X)$?

The Uniform Distribution

Let X be a continuous **uniform random variable**.

The probability density function of X is:

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise}, \end{cases}$$

where $[a, b]$ is the interval on which X is defined.

The Uniform Distribution

We write

$$X \sim \mathrm{Uniform}(a, b)$$

to mean that X is drawn from a uniform distribution on an interval $[a, b]$.

The Uniform Distribution

The mean of a uniformly distributed random variable is:

$$E(X) = \frac{a+b}{2}.$$

Note that this is the midpoint of the interval on which X is defined.

The variance of a uniformly distributed random variable is:

$$V(X) = \frac{(b-a)^2}{12}.$$

Example: The Uniform Distribution

Example: Suppose a train has a uniform probability of arriving between 0 and 10 minutes late.

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6. What is the variance of amount of time a train will arrive past the scheduled time?

The Exponential Distribution

Let X be an **exponential random variable**.

The probability density function of X is:

$$f_X(x) = \begin{cases} \frac{1}{\beta} e^{-\frac{x}{\beta}}, & x \geq 0, \\ 0, & \text{otherwise,} \end{cases}$$

where $\beta > 0$ is a parameter.

The Exponential Distribution

We write

$$X \sim \text{Exponential}(\beta)$$

to mean that X is drawn from an exponential distribution of parameter β.

The mean of a exponentially distributed random variable is:

$$E(X) = \beta.$$

The variance of a uniformly distributed random variable is:

$$V(X) = \beta^2.$$

Example: The Exponential Distribution

Example: Suppose a train is always late but the distribution of (late) arrival times is given by an exponential distribution with $\beta = 5$. What is the probability a train will arrive between 0 and three minutes late?

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